

## CS-844 Semester Project

**Due Date May 3rd, 2026 11:59PM**

**Deliverables:** Please upload the following on LMS by the deadline:

1. A pdf of the reports on LMS.
2. Your code. It should contain a pdf of jupyter notebook or console showing the output of your code.

**Code Demo and Viva:** You have to come to my office and demonstrate your code on May 4<sup>th</sup>, 2026. I will be available all day from 9.30am to 6pm. Please coordinate with me on Whatsapp and book your group's 15 minute slot. Slots will be on first come first serve basis, so make sure to reach out as soon as possible.

**Presentation:** Prepare a 10 minute presentation for the paper you did **NOT** implement. Presentations will be held from 5.30pm to 9pm on May 6<sup>th</sup>, 2026. Everyone is required to be present for all presentations.

**Groups:** Groups of at most 3 people need to be formed. You can do the project individually as well but there is no extra grade for that. Please submit your groups using this link:

<https://forms.gle/UywStiQ1qH8Mj48u8>

**Evaluation:** Grading will be done on an **individual** basis.

Viva and Demo – 50%

Report – 30%

Presentation – 20%

## **Description:**

This is primarily an open-ended research project. The main purpose is for you to spend some time doing literature review and gain a deeper understanding of specific Generative Deep Models. You can use components of your ongoing research if applicable.

## **Literature Review:**

You have to select **TWO** research papers with the following constraints:

1. The papers should be relatively recent. No more than 6 years old.
2. The papers should be published in well known machine learning venues such as NeurIPS, ICML, ICLR, CVPR, JMLR, TPAMI etc. or have a significant number (>100) citations.
3. The papers should be related to Generative Deep Models.

**Please submit your group's choice of the two papers by April 15<sup>th</sup>, 2026** using the following link: <https://forms.gle/KM9ysgsUqQseQVUX7>

## **Report:**

Once you have selected the two papers you have to write a 3 to 4 page review of each paper covering the following:

1. A short but comprehensive summary of the paper.
2. What kind of Generative Deep Model is used / proposed in the paper.
3. What makes the proposed method better than the previous available methods. Especially what makes it better than the basic versions of the same type of models.
4. What are the downsides of the proposed method.
5. What improvements can be made (you can look at papers published after the selected paper for help).

## **Implementation:**

After that, you will have to choose one of these papers and implement them yourself. Be mindful of resource constraints that you might run into. E.g. if you are working with a method that generates images, you can consider implementing a simpler/smaller version of it and train it on simpler datasets like MNIST / CIFAR10 etc.